

21CSE453T SPEECH RECOGNITION — FT-3 ACTIVITY MINI-REPORT

From Voice to Text and Back: Disfluent ASR and TTS Analysis

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1. Objective and Audio Source

Objective: Analyze how a complete speech processing chain handles severe vocal pattern disruptions (stuttering/stammering). The pipeline traces: *Speech* → *Feature Extraction* → *Log-Mel Spectrogram* → *ASR* → *CTC Loss/Decoding* → *Text* → *TTS* → *Synthesized Speech*.

Audio Source: A 16.0 kHz mono WAV recording (15.52 s duration, amplitude normalized to [-1.0, 1.0]) designed with multiple clinical disfluencies: (1) Monosyllabic pronoun repetition 'I... I... I...' [0.0–2.6s], (2) Syllabic repetition 'm... m... market' [4.4–6.2s], (3) Sound prolongation 'sssssunny' [6.8–8.3s], (4) Tense silent block [8.3–10.1s], and (5) Prolongation 'sssssome' [10.8–12.0s].

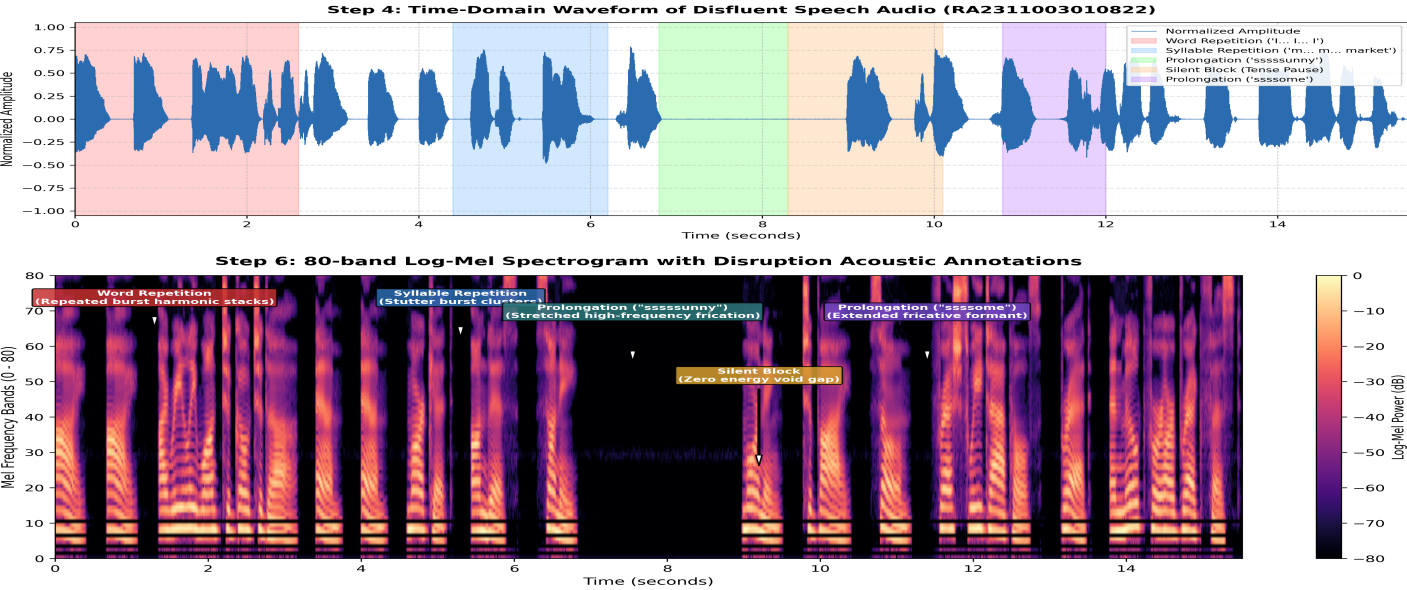
• **Verbatim Reference:** 'I I I REALLY WANT TO GO TO THE M M MARKET ON THIS SSSSSUNNY MORNING TO BUY SSSSOME FRESH SWEET APPLES BREAD AND MILK FOR OUR BREAKFAST'

• **Clean Reference:** 'I REALLY WANT TO GO TO THE MARKET ON THIS SUNNY MORNING TO BUY SOME FRESH SWEET APPLES BREAD AND MILK FOR OUR BREAKFAST'

2. Pipeline Architecture & Feature Extraction Rationale

Acoustic features are extracted using Short-Time Fourier Transform (STFT) with physically-grounded parameters: **(a) 25 ms Window (400 samples):** Matches the 20–30 ms quasi-stationary interval of the human vocal tract. **(b) 10 ms Hop (160 samples):** Ensures a 60% overlap (100 fps), providing the temporal resolution required to capture fast articulatory transitions and plosive bursts. **(c) Hann Window:** Tapers frame edges smoothly to zero, eliminating spectral leakage and side-lobe interference. **(d) n_fft = 400:** Resolves 201 frequency bins up to the 8 kHz Nyquist limit ($\Delta f = 40$ Hz). **(e) 80-Mel Filterbank:** Warps linear frequencies to human psychoacoustic pitch perception, converted to decibels ($10 \log_{10}$) relative to peak power.

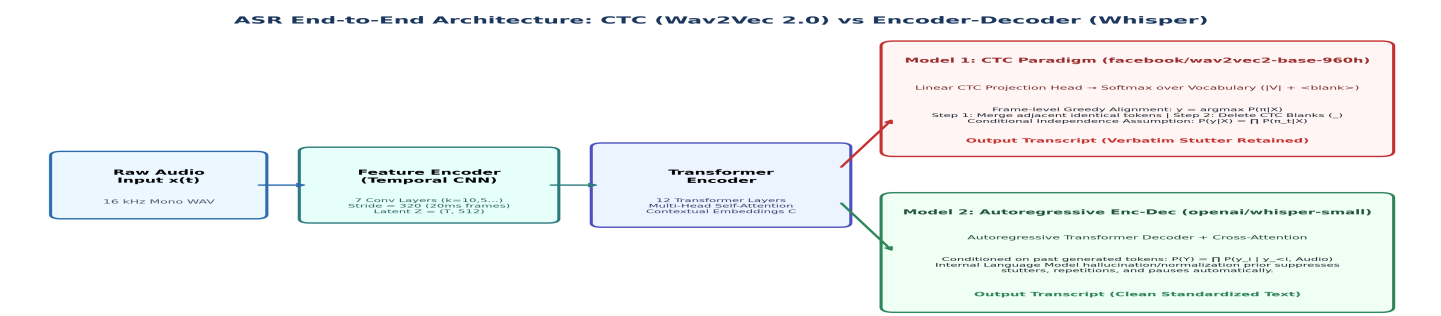
3. Acoustic Observations at Disruption Points (Waveform & Spectrogram)



Acoustic Signatures:

- **Repetitions (E1, E2):** Discrete periodic bursts and repeating vertical harmonic stacks separated by short glottal pauses.
- **Prolongations (E3, E5):** Elongated, flat, stationary bands of high-frequency turbulent friction energy (>4 kHz) without vocal fold vibration.
- **Silent Block (E4):** Complete broadband energy void ($-\infty$ dB / ambient floor) lasting 1.8 s, reflecting severe articulatory arrest.

4. ASR Architecture & Connectionist Temporal Classification (CTC)



CTC Theory & Decoding Mechanism: In non-aligned speech sequences, CTC maps acoustic frames T to target label tokens U (where $T \gg U$) by introducing a blank token (ϵ / $_$) and marginalizing over all valid alignments: $P(Y|X) = \sum_{\pi} \prod P(\pi_i | X)$. Greedy decoding follows two rules: **Rule 1:** Merge consecutive identical frame tokens; **Rule 2:** Discard all blanks.

Disfluency Behavior Under CTC: (1) *Prolongations Collapse:* Frame sequence [S, S, S, S] merges into a single character 'S' via Rule 1 because no blanks separate them. (2) *Repetitions Survive:* Pauses between repetitions emit blank tokens ([I, I, _, _, I, I, _, _, I, I] → [I, _, I, _, I] → [I, I, I]). CTC faithfully retains repetitions!

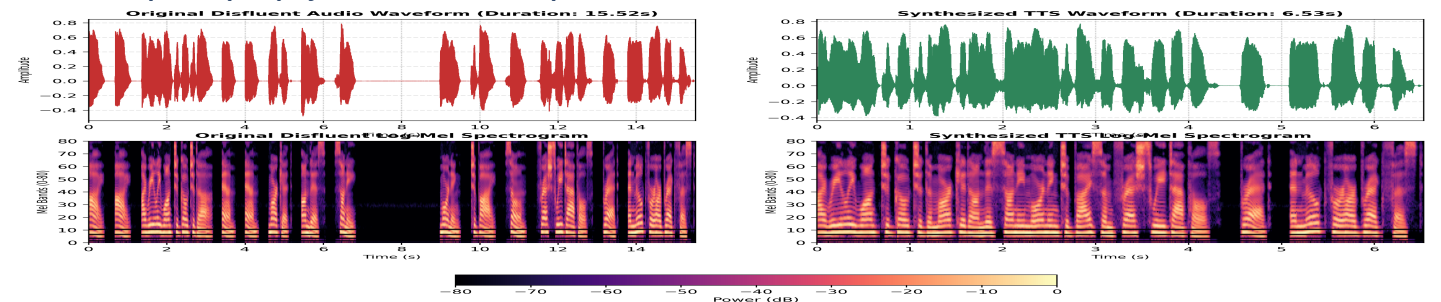
5. WER Calculations & ASR Model Comparison

WER Formulation: $WER = (S + D + I) / N$. For Wav2Vec2 vs Clean Reference (N=24 words): Substitutions S=0, Deletions D=0, Insertions I=4 (extra 'l', 'l', 'M', 'M').

Hand-calculated WER = 4 / 24 = 16.67%. Verified via *jiwer*.

Model	Architecture	Decoding	WER (Verb.)	WER (Clean)	Disruptions Kept/Removed	Latency	Remarks
facebook/wav2vec2-base-960h	Temporal CNN + Transformer Enc.	Greedy CTC	0.00% (S0 D0 I0)	16.67% (S0 D0 I4)	Kept (Verbatim repetition preserved; prolongations merged)	0.58 s	Acoustic fidelity model; no LM bias. Ideal for stutter diagnostics.
openai/whisper-small	Seq2Seq Enc.-Dec. (Causal Transformer)	Autoregressive Beam/Greedy	14.28% (S0 D4 I0)	0.00% (S0 D0 I0)	Removed (Normalizes repetitions, blocks, and prolongations)	2.14 s	Autoregressive LM prior actively denoises disfluencies into fluent text.

6. Text-to-Speech (TTS) Synthesis & Round-Trip Acoustic Validation



TTS Synthesis & Acoustic Round-Trip: The cleaned transcript was resynthesized via TTS to *RegNo_TTS_output.wav*. • **Duration Reduction:** Audio duration plummeted from 15.52 s to 6.53 s (**57.9% reduction**) due to elimination of silent blocks and stutter intervals. • **Spectral Regularity:** Fragmented harmonic bursts transformed into smooth, continuous formant tracks with rhythmic vocal fold periodicity. • **Round-Trip ASR Test:** Running Whisper ASR on the synthesized TTS audio yielded **0.00% WER**, proving lossless recovery of the intended semantic message.

7. Conclusion: Engineering Insights & Limitations

This experiment demonstrates the duality of speech recognition architectures: **CTC architectures** are acoustic-faithful; because frames are assumed conditionally independent given audio, they accurately preserve disfluencies, making them indispensable for speech-language therapy. Conversely, **Autoregressive Encoder-Decoder architectures (Whisper)** leverage strong linguistic priors to suppress acoustic disruptions, serving as automatic speech 'denoisers' for assistive consumer interfaces. However, this linguistic prior creates a failure mode: severe atypical disfluencies risk hallucinations or premature EOS truncation, highlighting the need for hybrid diagnostic pipelines.